

Alex Shuey

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EDUCATION

Cornell University, College of Engineering, Ithaca, NY

Bachelor of Science in Mechanical Engineering May 2027, Masters in Aerospace Engineering Dec 2027

TECHNICAL EXPERIENCE

Cornell University Design Build Fly

November 2023 – Present

Senior Advisor, August 2026 - Present | Propulsion Sub-Team Lead, August 2025 - August 2026 | Propulsion Sub-Team Member, November 2023 - August 2025

- Led 6 mechanical and electrical engineers through propulsion design, manufacturing, testing, and aircraft integration
- Directed propulsion testing and design reviews, using structured test procedures and documentation to drive design decisions and manage technical risk
- Performed propulsion analysis and validation using MATLAB, Python, eCalc, static thrust testing, wiring, and aircraft integration

Cornell University Sibley School of Mechanical and Aerospace Engineering

June 2026 - Present

Instructional Laboratories Intern

- Designed and fabricated hardware for undergraduate system dynamics experiments, including CAD-designed and 3D-printed adapters integrating motors, tachometers, and potentiometers into rotating-system test setups
- Built and integrated experimental instrumentation by soldering 200 BNC connectors, assembling custom DAQ hardware using NI USB-6001 systems, and wiring an additional laboratory test station
- Supported development and maintenance of MAE 3260 experiments through PCB soldering, fabrication, troubleshooting, and iterative prototype testing

Cornell University Duffield College of Engineering

May 2025 - May 2026

Student Researcher

- Collaborated within a 3-person engineering team to support the development of a Magnus-effect ultralight VTOL aircraft, coordinated design, fabrication, and testing efforts across project phases
- Tracked and executed testing plans and performance milestones, including flight tests achieving ten five-minute hovers with 4 kg payload (25% above target)
- Utilized SOLIDWORKS, ANSYS, and MATLAB to support design validation and ensure performance requirements were met within project constraints.

Cornell University Sibley School of Mechanical and Aerospace Engineering

August 2026 – Present

Fluid Mechanics Teaching Assistant

- Hold office hours, lead discussion sections and lab activities, answer student questions.
- Review homework, quizzes, exams, and solutions for accuracy; assist with course logistics.

Cornell University Sibley School of Mechanical and Aerospace Engineering

August 2026 – Present

Introduction to Aeronautics Grader

- Grade and review homework, quizzes, and exams for accuracy and consistency; assist with course logistics and solution verification.

LEADERSHIP & COMMUNITY INVOLVEMENT

Cornell University Office of Visitor Relations

May 2024 – Present

Information Specialist (Tour Guide)

- Selected to represent Cornell University as a front-facing ambassador, lead 75-minute, engaging, high-energy tours while walking backwards across campus, maintaining group safety and attention through humor, storytelling, and real-time audience engagement.
- Provide high-level customer service at the front desk and operate the university switchboard, managing high call volume and competing priorities with professionalism

SKILLS & ACTIVITIES

Skills: SOLIDWORKS, Fusion 360, ANSYS, MATLAB, Python, eCalc, Arduino, soldering, PCB assembly, machining (mill, lathe), aviation trainee

Other Activities: Society of Women Engineers - Outreach Committee Coordinator; Cornell Engineering Ambassador; Women of Aeronautics and Astronautics, Delta Delta Delta Sorority